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(54) **SAWN INDIVIDUALLY MOLDED QUAD FLAT NO-LEAD (QFN) SEMICONDUCTOR PACKAGE**

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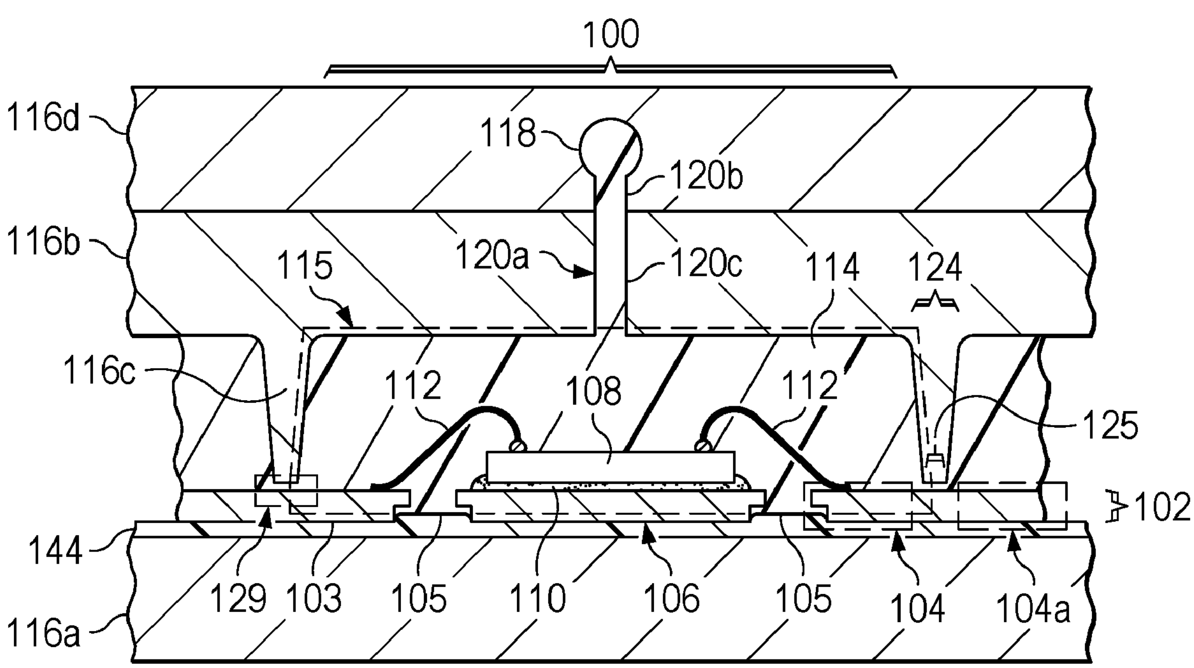
(57) **ABSTRACT**

A semiconductor package may have a microelectronic component electrically coupled to a plurality of leads which extend to a perimeter of the semiconductor package. A mold compound, which is electrically insulating, contacts the plurality of leads and the microelectronic component. Mold Array Process (MAP) molded Quad Flat No-Lead (QFN) packaging may apply mold compound to multiple microelectronic components simultaneously. After molding, the packages are cured and singulated. A mold compound middle mold chase with a mold cavity for each package and a mold compound injection port associated with each semiconductor package to distribute the mold compound may decreases micro voiding of the mold compound, raise the coefficient of thermal expansion, and improve board level reliability. Semiconductor packages formed with a mold compound injection port corresponding to each package may have a gate cull on the exterior of the semiconductor package where the mold compound enters the mold cavity.



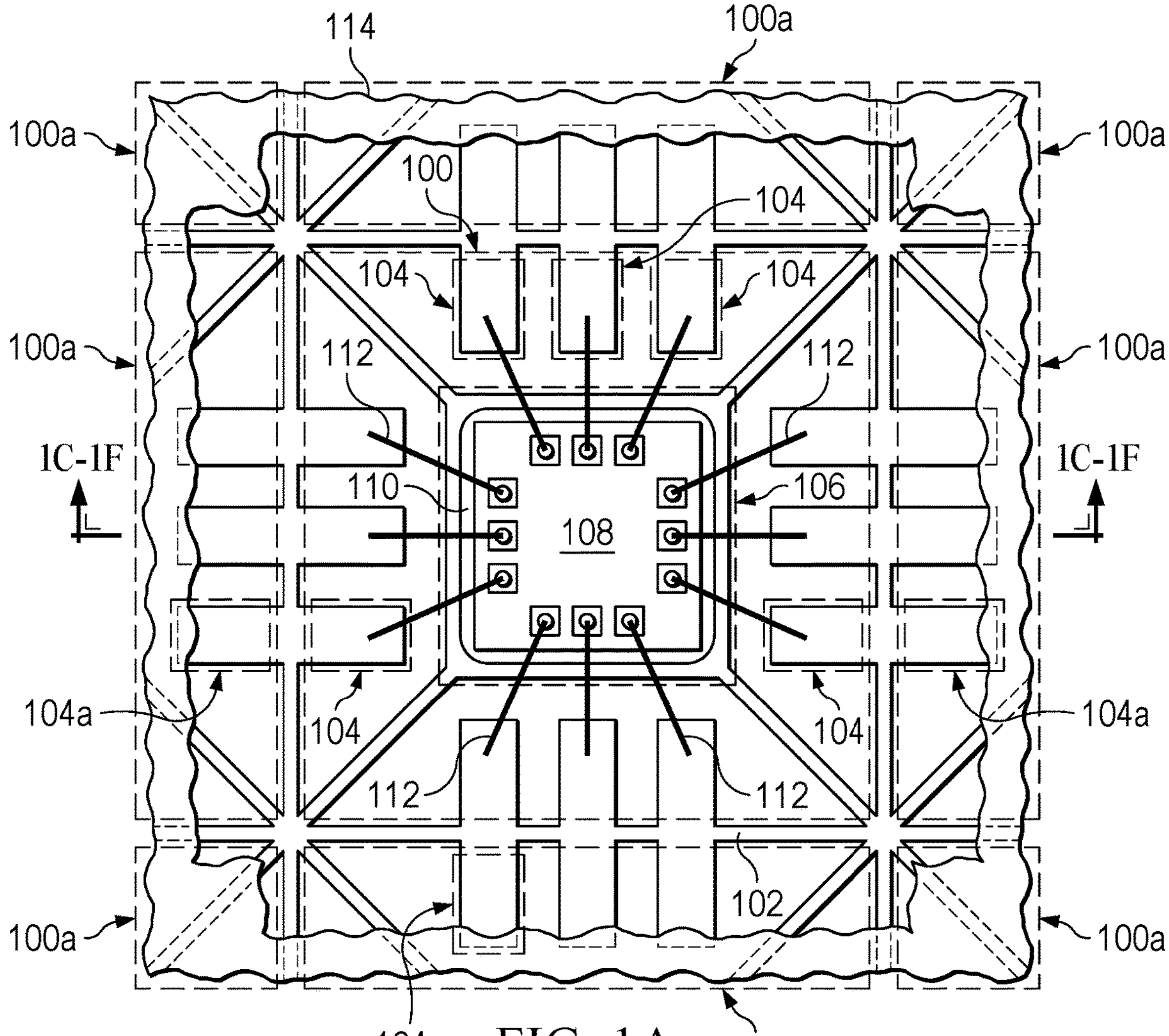

114
100a
100a
100a
100
104
104
104
100a
100a
112
112
1C-1F
1C-1F
110
106
108
104a
104
104
104a
112
112
102
100a
100a
104a
100a

FIG. 1A

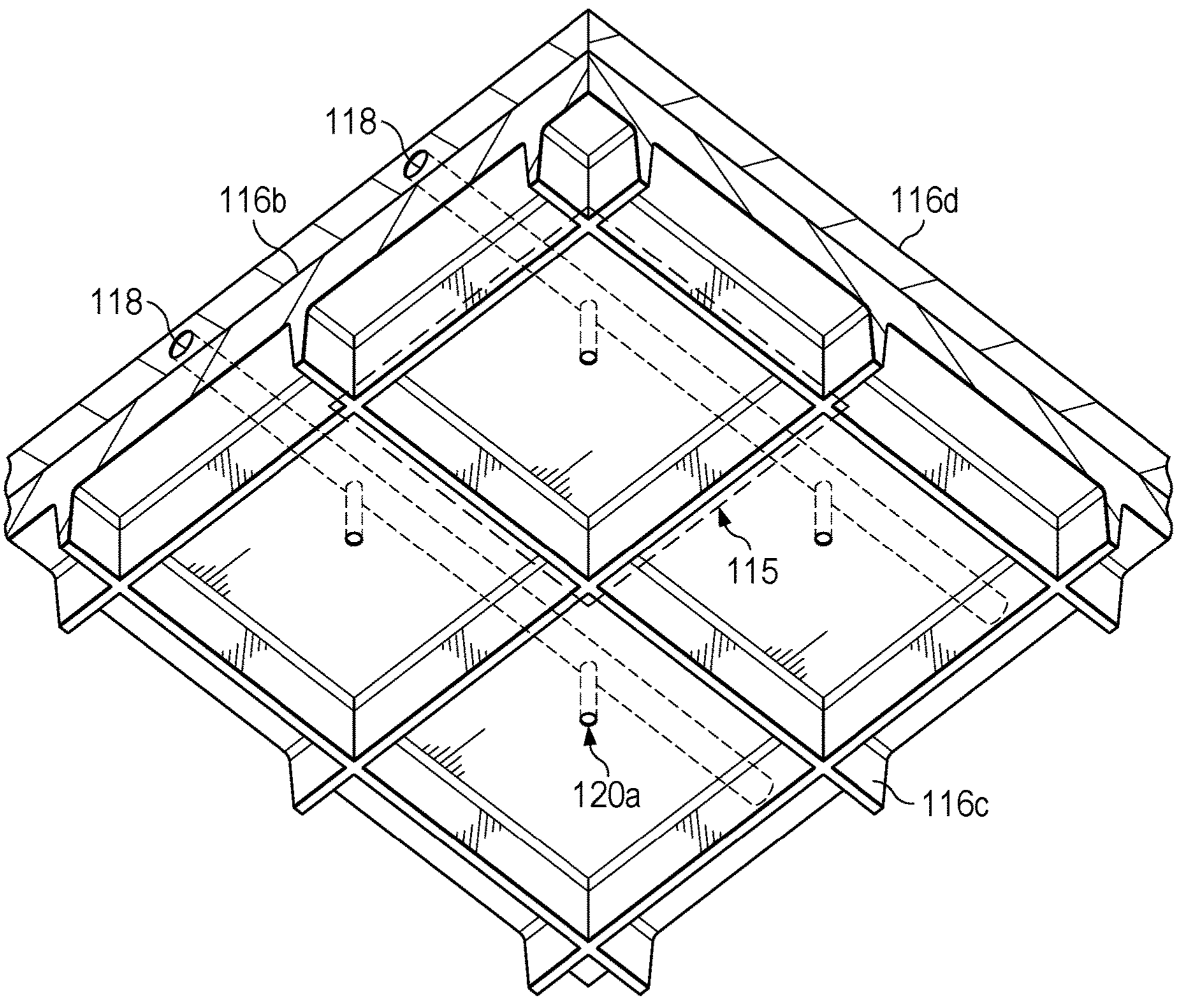


FIG. 1B

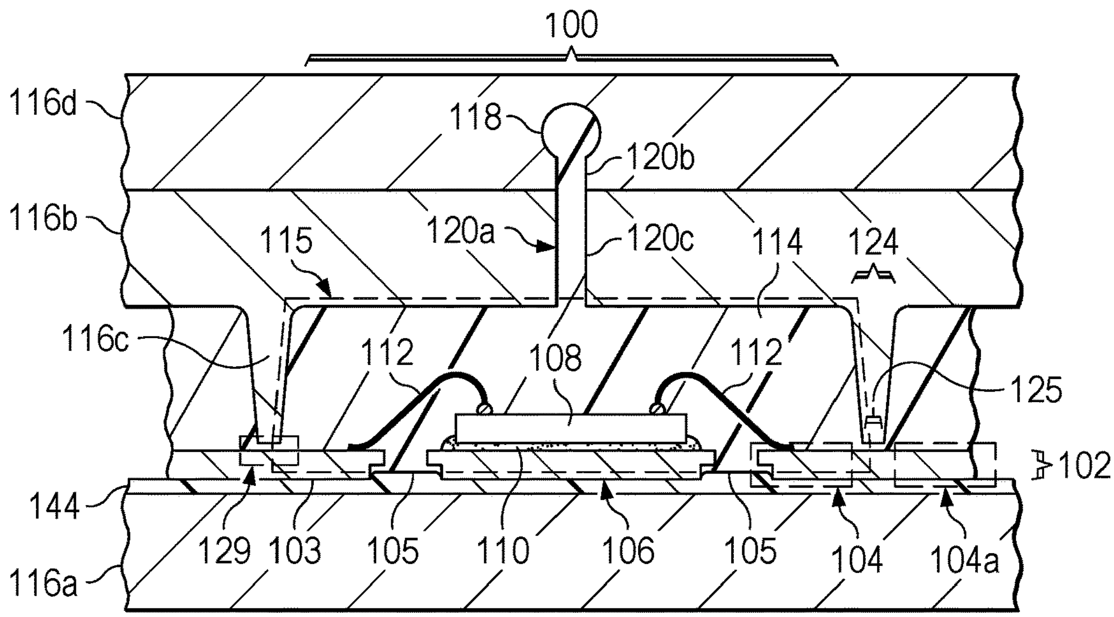

100
116d
118
120b
116b
120a
120c
114
124
115
116c
112
108
112
125
102
144
129
103
105
110
106
105
104
104a
116a

FIG. 1C

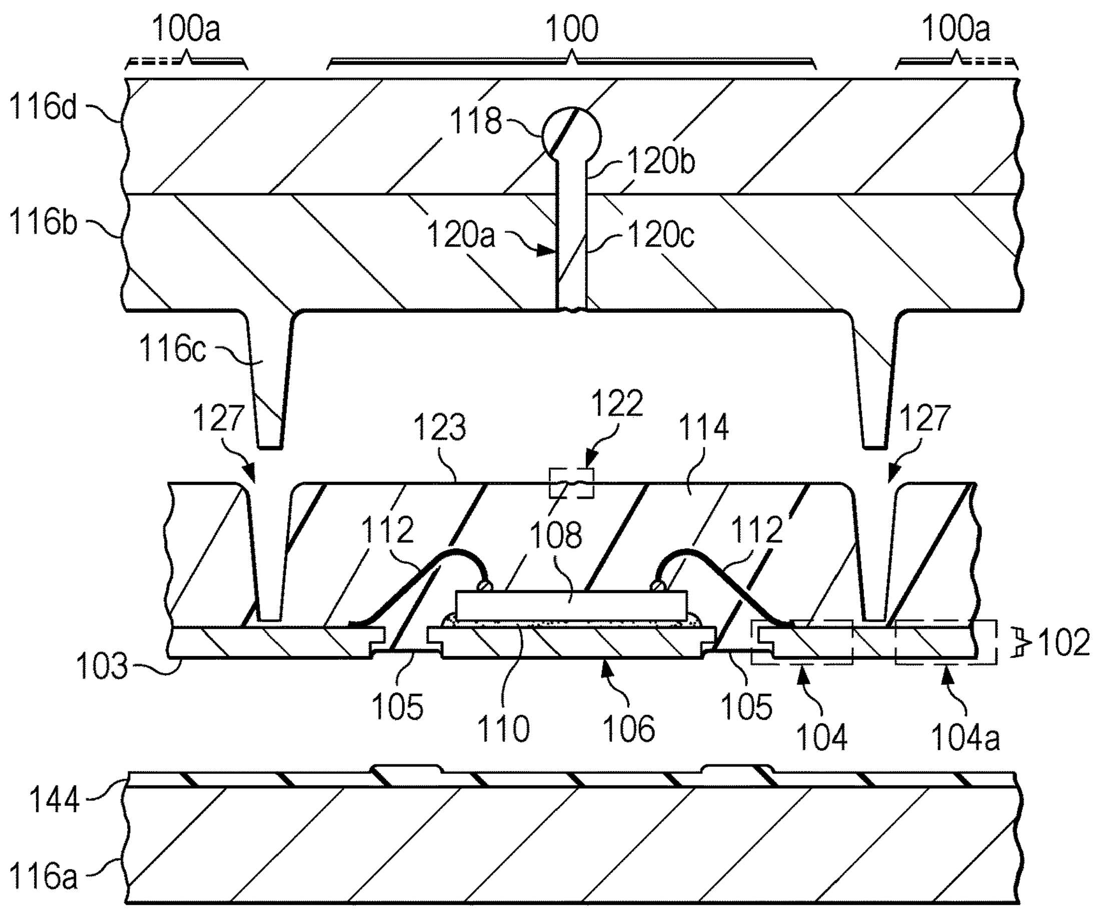

100a
100
100a
116d
118
120b
116b
120a
120c
116c
127
123
122
114
127
112
108
112
102
103
105
110
106
105
104
104a
144
116a

FIG. 1D

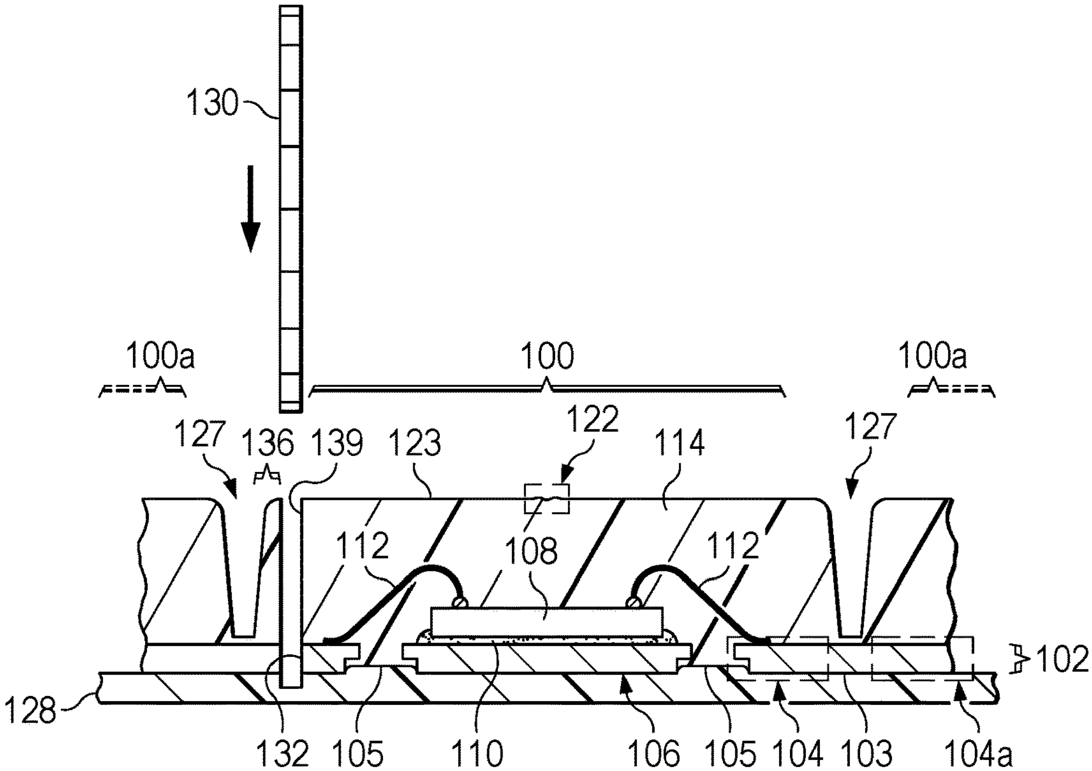

130
100a
100
100a
127
136
139
123
122
114
127
112
108
112
102
128
132
105
110
106
105
104
103
104a

FIG. 1E

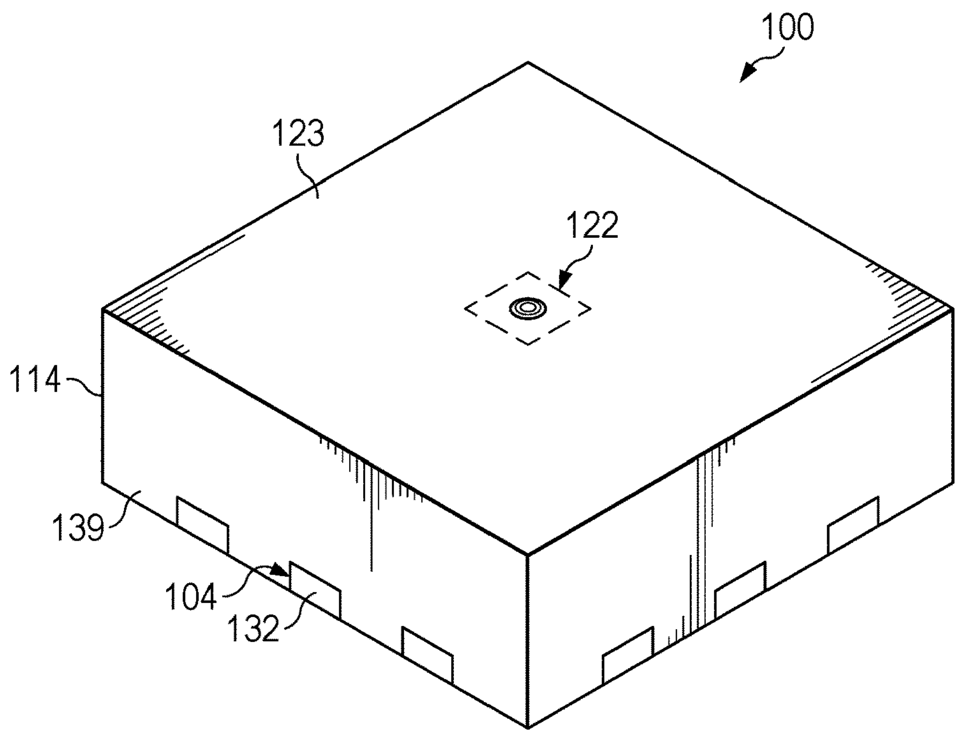

100
123
122
114
139
104
132

FIG. 1F

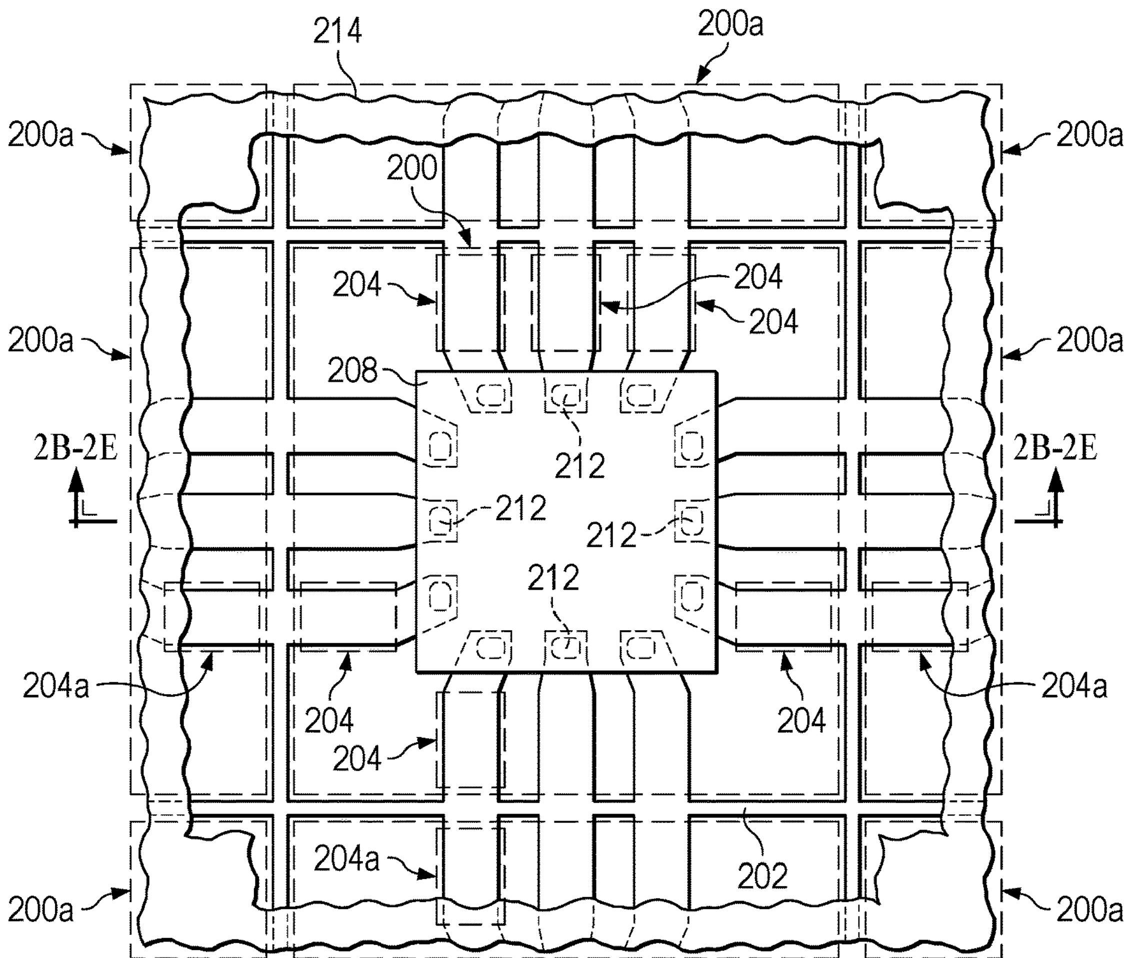

214
200a
200a
200a
200
204
204
204
200a
200a
208
2B-2E
212
2B-2E
212
212
212
204a
204a
204
204
204
204a
202
200a
200a
200a

FIG. 2A

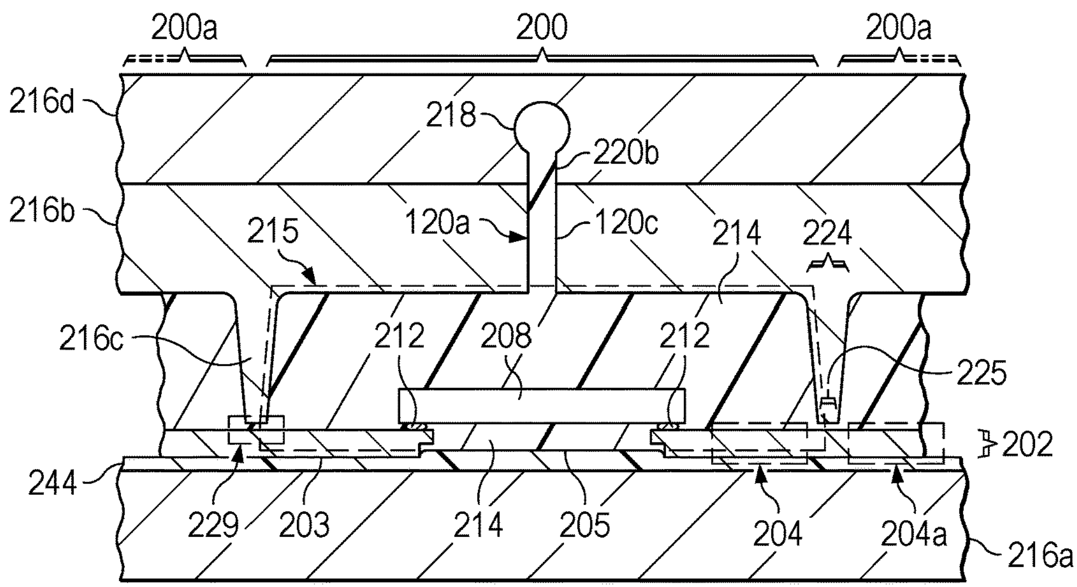

200a
200
200a
216d
218
220b
216b
215
120a
120c
214
224
216c
212
208
212
225
202
244
229
203
214
205
204
204a
216a

FIG. 2B

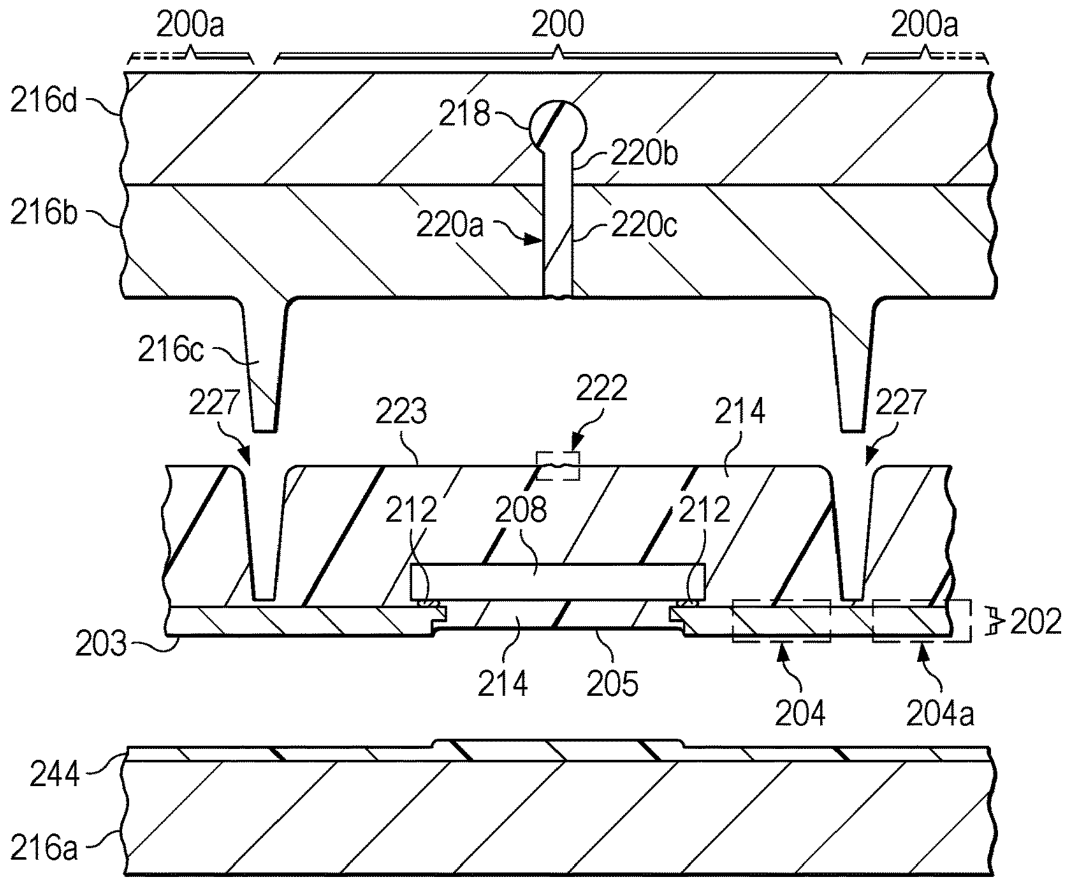

200a
200
200a
216d
218
220b
216b
220a
220c
216c
227
223
222
214
227
212
208
212
202
203
214
205
204
204a
244
216a

FIG. 2C

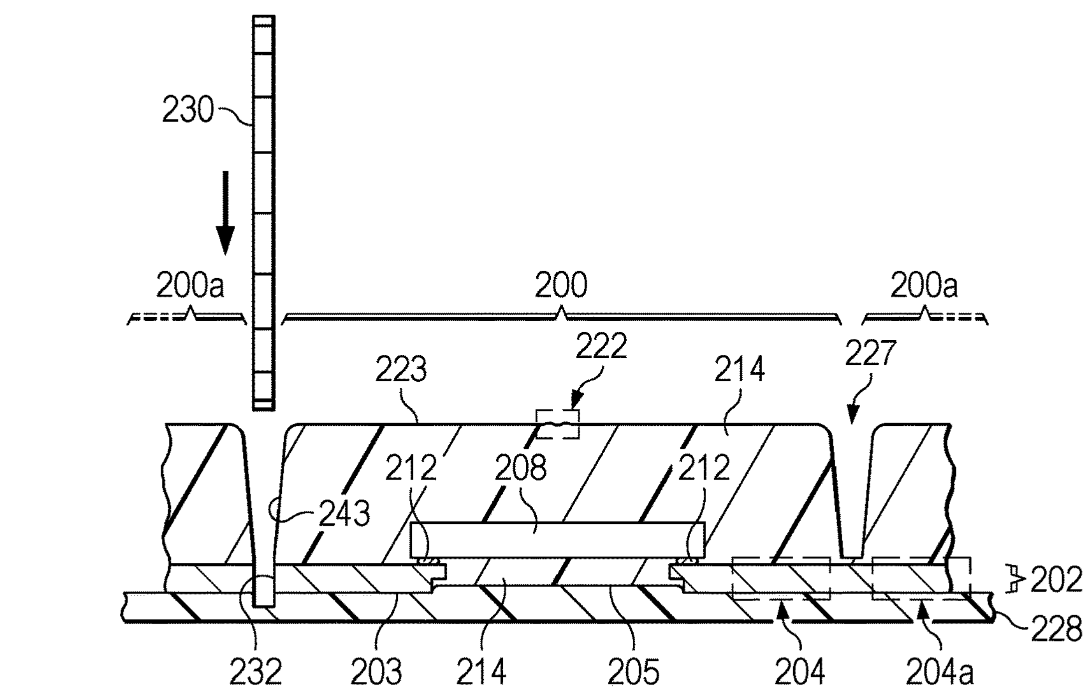

230
200a
200
200a
223
222
214
227
212
208
212
243
202
228
232
203
214
205
204
204a

FIG. 2D

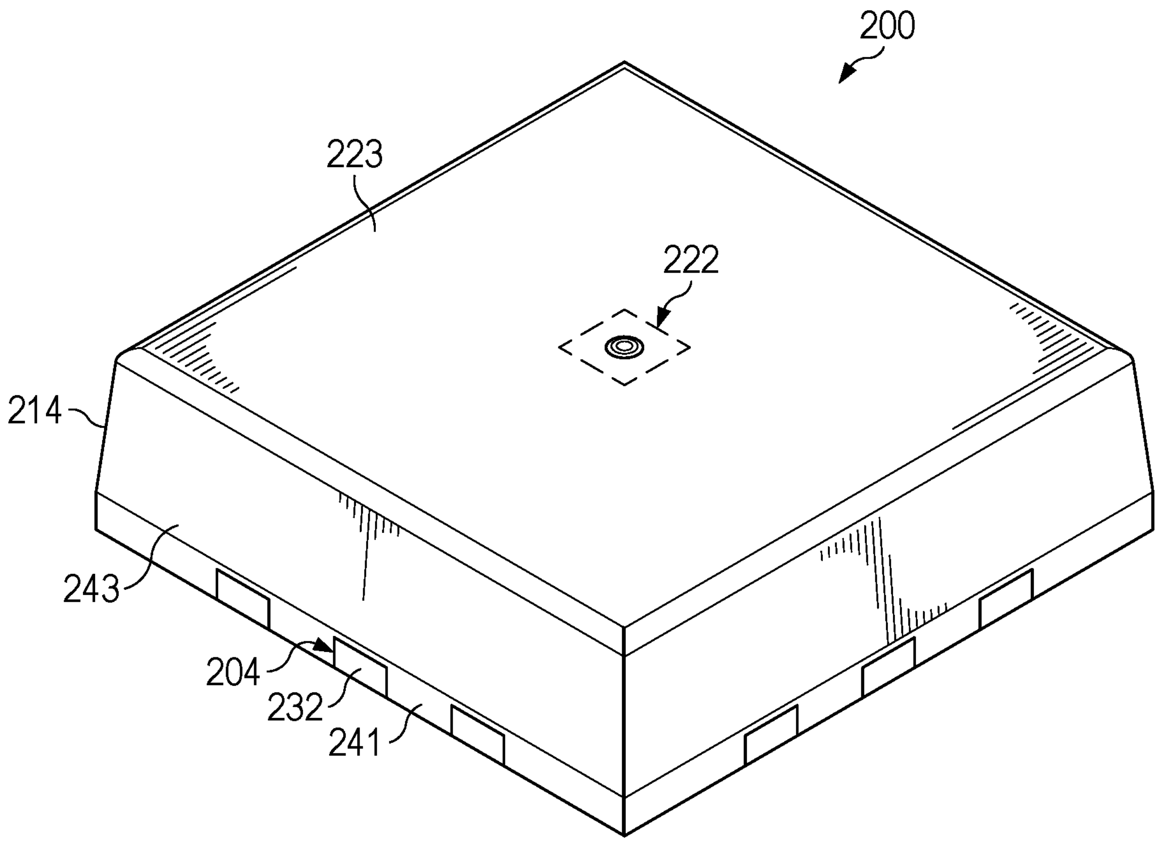


FIG. 2E

SAWN INDIVIDUALLY MOLDED QUAD FLAT NO-LEAD (QFN) SEMICONDUCTOR PACKAGE

TECHNICAL FIELD

[0001] This disclosure relates to the field of semiconductor packages. More particularly, but not exclusively, this disclosure relates to quad flat no-lead semiconductor packages.

BACKGROUND

[0002] Quad flat no-lead (QFN) packages have emerged as a popular choice for compact and high-performance microelectronic component packaging. QFN packages have metal pads on the bottom surface of the package, enabling a higher pin density in a smaller form factor. QFN packages exhibit shorter electrical paths, leading to lower inductance, reduced parasitic capacitance, and enhanced high-frequency performance. QFN packages offer simplified assembly processes, increased automation compatibility, and reduced material costs.

SUMMARY

[0003] A semiconductor package may have a plurality of leads around a perimeter of the semiconductor package and a microelectronic component electrically coupled to the plurality of the leads. The plurality of leads may be part of a lead frame. A mold compound, which is electrically insulating, contacts the leads and the microelectronic component. Mold Array Process (MAP) molded Quad Flat No-Lead (QFN) packaging may apply mold compound to multiple microelectronic components and their associated leads simultaneously in one large, common mold cavity. After molding, the semiconductor packages are then cured and singulated by a sawing method. Semiconductor packages formed by MAP molded QFN packages may have a lower coefficient of thermal expansion and a wider variation of coefficient of thermal expansion than punched individually molded single die QFN packages formed using the same mold compound. A lower coefficient of thermal expansion and a wider variation of coefficient of thermal expansion may result in degraded solder joint and board level reliability for the MAP molded QFN packages. The difference in coefficient of thermal expansion may be due to higher levels of micro voiding in the mold compound during polymerization of the mold compound during a mold compound curing process. A mold compound middle mold chase with a mold cavity for each semiconductor package and a mold compound injection port associated with each semiconductor package may decrease micro voiding, yielding mechanical properties similar to the mold compound physical bulk properties which may improve solder joint and board level reliability. A mold compound injection port over the center point of the mold cavity may minimize wire sweep. Semiconductor packages formed with a mold compound injection port associated with each semiconductor package may have a gate cull mark on the exterior of the semiconductor package at the site where the mold compound enters the mold cavity.

BRIEF DESCRIPTION OF THE FIGURES

[0004] FIG. **1**A is a top-down view of an example semiconductor package with a semiconductor component on a semiconductor package lead frame.

[0005] FIG. **1**B is a perspective view of an example semiconductor package upper mold chase and middle mold chase. The middle mold chase includes mold cavities and a mold compound injection port for each package of the middle mold chase.

[0006] FIGS. **1**C-**1**F are cross sections of an example semiconductor package in various states of formation.

[0007] FIGS. **2**A-**2**E are cross sections of an example semiconductor package in various states of formation.

DETAILED DESCRIPTION

[0008] The present disclosure is described with reference to the attached figures. The figures are not drawn to scale and they are provided merely to illustrate the disclosure. Several aspects of the disclosure are described below with reference to example applications for illustration. It should be understood that numerous specific details, relationships, and methods are set forth to provide an understanding of the disclosure. The present disclosure is not limited by the illustrated ordering of acts or events, as some acts may occur in different orders and/or concurrently with other acts or events. Furthermore, not all illustrated acts or events are required to implement a methodology in accordance with the present disclosure.

[0009] In addition, although some of the examples illustrated herein are shown in two dimensional views with various regions having depth and width, it should be clearly understood that these regions are illustrations of only a portion of a device that is actually a three-dimensional structure. Accordingly, these regions will have three dimensions, including length, width, and depth, when fabricated on an actual device. Moreover, while the present disclosure may be illustrated by examples directed to active devices, it is not intended that these illustrations be a limitation on the scope or applicability of the present disclosure. It is not intended that the active devices of the present disclosure be limited to the physical structures illustrated. These structures are included to demonstrate the utility and application of the present disclosure to various examples.

[0010] It is noted that terms such as top, bottom, over, above, and under may be used in this disclosure. These terms should not be construed as limiting the position or orientation of a structure or element, but should be used to provide spatial relationship between structures or elements. The terms "lateral" and "laterally" refer to directions parallel to a plane corresponding to a surface of a layer, for example a top surface of a semiconductor substrate. Moreover, the term "approximately," as used herein, may refer to ±5% to ±10% variations of the recited values in some cases. In other cases, the term "approximately" may refer to ±10% to ±20% variations of the recited values. Microelectronic devices are being continually improved to reliably operate with higher performance and smaller feature sizes.

[0011] A semiconductor package contains a microelectronic component electrically coupled to a plurality of leads around a perimeter of the semiconductor package. The plurality of leads may be attached to a lead frame. The lead frame contains a plurality of semiconductor packages. A mold compound, which is electrically insulating, is formed on the plurality of leads and the microelectronic component. The mold compound is formed between a bottom mold chase and a middle mold chase. A top mold chase is on the middle mold chase. It is advantageous for the middle mold chase to have a series of middle mold chase ridges which form a portion of a mold cavity surrounding each microelectronic component to be packaged and for each mold cavity to have a mold compound injection port to deliver the mold compound to each mold cavity, the mold compound being distributed to each mold compound injection port by mold compound runners within the top mold chase. The ability to deliver mold compound to each individual semiconductor package is advantageous as it may minimize micro voiding of the mold compound which may lead to a lower coefficient of thermal expansion which may lead to solder bond and board level reliability degradation compared to a semiconductor package formed with a middle mold chase without middle mold chase ridges and/or a mold compound injection port for each semiconductor package. After the formation of the mold compound, the middle mold chase and the bottom mold chase are removed. When the middle mold chase is removed, a gate cull mark is left on the semiconductor package where the mold compound detaches from the top mold compound injection port.

[0012] After the formation of the semiconductor package on the lead frame, individual semiconductor packages may be singulated by sawing through the leads. The sidewalls of the singulated packages may be vertical with respect to the top surface of the semiconductor package if the semiconductor package is sawn between the microelectronic component and a mold compound depression formed by the middle mold chase ridges between semiconductor packages. A tapered sidewall profile may result if the singulation sawing process is along the mold compound depression. Singulation by sawing enables forming a plurality of semiconductor packages concurrently using the lead frame, as the mold compound can be formed on all the semiconductor packages in a single operation.

[0013] It is noted that terms such as top, bottom, over, under, and below may be used in this disclosure. These terms should not be construed as limiting the position or orientation of a structure or element, but should be used to provide spatial relationship between structures or elements.

[0014] FIG. **1**A is a top-down view of a semiconductor package **100** including a microelectronic component **108** attached to a lead frame **102**. FIG. **1**B is a perspective view of a middle mold chase **116***b* and a top mold chase **116***d*. FIG. **1**C through FIG. **1**E are example cross sections during the formation of the semiconductor package **100**.

[0015] FIG. **1**A is a top-down view of the semiconductor package **100** at the first stage of formation. Referring to FIG. **1**A, the semiconductor package **100** may be formed using a lead frame **102**. The lead frame **102** of this example accommodates additional semiconductor packages **100***a* adjacent to the semiconductor package **100**.

[0016] The semiconductor package **100** includes a plurality of leads **104**. Similarly, the additional semiconductor packages **100***a* also include a plurality of additional leads **104***a*. The semiconductor package **100** of this example includes a die pad **106** connected to the lead frame **102**.

[0017] A microelectronic component **108** is attached to the die pad **106**. The microelectronic component **108** may be manifested as an integrated circuit, a discrete component such as a power transistor, a passive component such as a transformer or a filter, a micro electromechanical system (MEMS) component, a sensor, an actuator, a microfluidic component, or an electro-optical component such as a micro-mirror array component, by way of example. The microelectronic component **108** may be attached to the die pad **106** by a die attach material **110**, such as solder, an electrically conductive adhesive, an electrically insulating adhesive, or a eutectic metal alloy.

[0018] Electrical connections **112** are formed between the microelectronic component **108** and the plurality of leads **104**, thus electrically coupling the microelectronic component to the plurality of leads **104**. The electrical connections **112** may be implemented as wire bonds, as depicted in FIG. **1**A and FIG. **1**C-FIG. **1**E. Alternatively, the electrical connections **112** may be implemented as ribbon bonds, clips, leads on resin coated copper, or solder bumps, by way of example.

[0019] Referring to FIG. **1**B, a perspective view is shown of a portion of the middle mold chase **116***b* and the top mold chase **116***d*. The middle mold chase **116***b* contains a grid of middle mold chase ridges **116***c* which are a part of the middle mold chase **116***b*, protruding from the middle mold chase **116***b* as an array, the middle mold chase ridges **116***c* forming a mold cavity **115** which surrounds each semiconductor package **100** during the mold compound formation process referred to in FIG. **1**C. The top mold chase **116***d* contains top mold chase runners **118** which allow the mold compound **114** (referred to in FIG. **1**C) to be distributed within the top mold chase **116***d*. The top mold chase runners **118** in the top mold chase **116***d* are in rows or columns over each row or column of semiconductor packages **100** of the lead frame **102**. The top mold chase runners **118** distribute mold compound to each semiconductor package **100** through a mold compound injection port **120***a*, the mold compound injection port **120***a* intersecting the top mold chase runners **118** and distributing the mold compound **114** into each mold cavity **115**. A first portion of the mold compound injection port **120***b* is in the top mold chase **116***d*, while a second portion of the mold compound injection port **120***c* is in the middle mold chase **116***b* (referred to in FIG. **1**C). While the example middle mold chase **116***b* shows one mold compound injection port **120***a* per mold cavity **115**, more than one mold compound injection port **120***a* per mold cavity is within the scope of the disclosure. Additionally, while mold compound injection port **120***a* of the example device is placed above the microelectronic component **108** of the semiconductor package **100** as referred to in FIG. **1**C, the top mold chase runners **118** and mold compound injection port **120***a* may be configured to allow the mold compound **114** to enter the mold cavity **115** through the middle mold chase ridges **116***c* i.e. From the side of the mold cavity **115**. It may be advantageous for the mold compound injection port **120***a* to be over a center point of the mold cavity **115** to minimize wire sweep of the electrical connections **112** during the injection of the mold compound **114** into the mold cavity **115**. (referred to in FIG. **1**C). The mold compound **114** is in a liquid state as it is injected into the mold cavity **115** through the mold compound injection port **120***a*. Wire sweep is a phenomenon where the electrical connections **112** may be deformed during the formation of the mold compound **114**.

[0020] Referring to FIG. **1**C, a cross section of the semiconductor package **100** is shown during the mold compound **114** formation. In FIG. **1**C, the lead frame **102** is placed between a bottom mold chase **116***a* and the middle mold chase **116***b*, with the top mold chase **116***d* on the middle mold chase **116***b*. In this example, a compressible mold relief film **144** is located between the bottom mold chase **116***a* and the bottom surface **103** of the plurality of leads

104. The compressible mold relief film **144** compresses between the bottom mold chase **116***a* and the plurality of leads **104**. After the injection of the mold compound **114** into the mold cavity **115**, the mold compound **114** may be flush with the bottom surface **103** of the plurality of leads **104** (not particularly shown) or may be recessed from the bottom surface **103** of the plurality of leads **104** such that the plurality of leads **104** extend below the mold compound bottom surface **105** as shown in FIG. **1**C. Middle mold chase ridges **116***c* form the mold cavity **115**. The top mold chase runners **118** provide a path for mold compound **114** to fill each the mold cavity **115** via the mold compound injection port **120***a*. The mold compound **114** encapsulates the plurality of leads **104**, the die pad **106**, the die attach material **110**, the microelectronic component **108** and the electrical connection **112** between the microelectronic component **108** and the plurality of leads **104**. The middle mold chase ridges **116***c* may have a chamfer with a middle mold chase ridge top width **124**, and a middle mold chase ridge bottom width **125**. The middle mold chase ridges **116***c* may be in contact with the plurality of leads **104**, or may have a middle mold chase ridge to lead space **129**. The mold compound **114** in this example is formed between the middle mold chase **116***b* and the compressible mold relief film **144** on the bottom mold chase **116***a*. The mold compound **114** is electrically insulating. The mold compound **114** may include epoxy, by way of example. The mold compound **114** may include filler particles, such as silicon dioxide particles or aluminum oxide particles, to adjust the mechanical properties of the mold compound **114**.

[0021] Referring to FIG. **1**D, a cross section is shown after the removal of the middle mold chase **116***b* and the bottom mold chase **116***a*. The removal of the middle mold chase **116***b* leaves a mold compound depression **127** resulting from the middle mold chase ridges **116***c* of the mold cavity **115**. During the removal of the middle mold chase **116***b*, a sprue (not specifically shown) in the mold compound injection port **120***a* breaks away from the mold compound **114** of the semiconductor package **100**. The breaking of the sprue from the semiconductor package **100** results in a gate cull mark **122** on the top surface **123** of the semiconductor package **100**. While the gate cull mark **122** of the semiconductor package **100** is on the top surface **123**, other locations of the gate cull mark **122** are within the scope of the disclosure, the location of the gate cull mark **122** being dependent of the location of the mold compound injection port **120***a*. The gate cull mark **122** may be circular in shape. Other shapes of the gate cull mark **122** are within the scope of the disclosure

[0022] FIG. **1**E is a cross section of the semiconductor package **100** of FIG. **1**D at a subsequent stage of formation during a singulation sawing process. Referring to FIG. **1**E, the semiconductor package **100** and the additional semiconductor packages **100***a* are mounted onto a saw tape ring **128**, which may also be referred to as a saw film. The semiconductor package **100** is singulated from the additional semiconductor packages **100***a* by a singulation sawing process using a singulation saw blade **130**. The sawing process may be between the edge of the microelectronic component **108** and the mold compound depression **127**.

[0023] In the example semiconductor package **100**, a saw blade to mold compound depression space **136** defines a space between the mold compound depression **127** and a path for the saw blade **130** such that the semiconductor package **100** when singulated has a mold compound sidewall **139** which is vertical with respect to the top surface **123** of the mold compound **114**. The mold compound sidewall **139** which is vertical results in a semiconductor package **100** after singulation with a perimeter of the mold compound **114** around the top surface **123** which is equal to the perimeter of the mold compound **114** around the bottom surface **103**.

[0024] The saw blade to mold compound depression space **136** may vary based on the size of microelectronic component **108** to be packaged, and allows the middle mold chase **116***b* to be used in a semiconductor package **100** formation process for microelectronic components **108** of multiple different sizes. The singulation sawing process cuts through and separates the plurality of leads **104** of the semiconductor package **100** and the plurality of additional leads **104***a* of the additional semiconductor packages **100***a*, as well as the mold compound **114** between the semiconductor package **100** and the additional semiconductor packages **100***a*. The semiconductor package **100** and the additional semiconductor packages **100***a* are subsequently removed from the saw tape ring **128**. The singulation sawing process produces exposed side faces **132** of the plurality of leads **104** which are free of striations, striations being typical of a semiconductor package **100** formed by shearing, instead of sawing the plurality of leads **104** from the plurality of additional leads **104***a* as shown in FIG. **1**E

[0025] FIG. **1**F is a perspective view of the semiconductor package **100** after singulation. The semiconductor package **100** after singulation may have a gate cull mark **122** and mold compound sidewalls **139**. The gate cull mark **122** may be on the exterior of the semiconductor package **100** on the top surface **123** of the mold compound **114**. Other features visible on the semiconductor package **100** after singulation include the plurality of leads **104** and exposed side faces **132** of the plurality of leads **104**.

[0026] FIG. **2**A is a top-down view of the semiconductor package **200** at the first stage of formation. Referring to FIG. **2**A, the semiconductor package **200** may be formed using a lead frame **202**. The lead frame **202** of this example may accommodates additional semiconductor packages **200***a* adjacent to the semiconductor package **200**. The semiconductor package **200** includes a plurality of leads **204**. Similarly, the additional semiconductor packages **200***a* also include a plurality of additional leads **204***a*.

[0027] A microelectronic component **208** is attached to the plurality of leads **204** through electrical connections **212**, i.e. Solder balls in the example semiconductor package **200**, thus electrically coupling the microelectronic component **208** to the plurality of leads **204**. Other method of electrical connection between the plurality of leads **204** and the microelectronic component **208** are within the scope of the disclosure. The microelectronic component **208** may be manifested as an integrated circuit, a discrete component such as a power transistor, a passive component such as a transformer or a filter, a MEMS component, a sensor, an actuator, a microfluidic component, or an electro-optical component such as a micro-mirror array component, by way of example.

[0028] Referring to FIG. **2**B, a cross section of a semiconductor package **200** shown in FIG. **2**A at the point in the packaging process when the mold compound **214** is formed. In FIG. **2**B, the microelectronic component **208** attached to the plurality of leads **204** is placed between a bottom mold chase **216***a* and the middle mold chase **216***b*. A top mold chase **216***d* is on the middle mold chase **216***b*. In this example, a compressible mold relief film **244** is located between the bottom mold chase **216***a* and the bottom surfaces **203** of the plurality of leads **204**. The compressible mold relief film **244** compresses between the bottom mold chase **216***a* and the plurality of leads **204**. After the injection of the mold compound **214** into the mold cavity **215**, the mold compound **214** may be flush with the bottom surface **203** of the plurality of leads **204** (not particularly shown) or may be recessed from the bottom surface **203** of the plurality of leads **204** such that the plurality of leads **204** extend below the mold compound bottom surface **205** as shown in FIG. **2**B. Middle mold chase ridges **216***c* which are part of the middle mold chase **216***b* and extend from the middle mold chase **216***b* to form the mold cavity **215** which surrounds the semiconductor package **200**. The top mold chase runners **218** distributes the mold compound **214** to each semiconductor package **200** through a mold compound injection port **220***a*, the mold compound injection port **220***a* intersecting the top mold chase runners **218** and distributing the mold compound **214** into each mold cavity **215**. A first portion of the mold compound injection port **220***b* is in the top mold chase **216***d*, while a second portion of the mold compound injection port **220***c* is in the middle mold chase **216***b*. The top mold chase runners **218** in the top mold chase **216***d* are in rows or columns over each row or column of microelectronic components **208** on the lead frame **202**. The mold compound **214** encapsulates the plurality of leads **204**, the microelectronic component **208** and the electrical connection **212** between the microelectronic component **208** and the plurality of leads **204**. The middle mold chase ridges **216***c* may have a chamfer with a middle mold chase ridge top width **224**, and a middle mold chase ridge bottom width **225**. The middle mold chase ridge **216***c* may be in contact with the plurality of leads **204**, or may have a middle mold chase ridge to lead space **229**. The mold compound **214** is electrically insulating. The mold compound **214** may include epoxy, by way of example. The mold compound **214** may include filler particles, such as silicon dioxide particles or aluminum oxide particles, to adjust the mechanical properties of the mold compound **214**.

[0029] Referring to FIG. **2**C, a cross section is shown after the removal of the middle mold chase **216***b* and the bottom mold chase **216***a*. The removal of the middle mold chase **216***b* leaves a mold compound depression **227** resulting from middle mold chase ridges **216***c*. During the removal of the middle mold chase **216***b*, a sprue (not specifically shown) in the mold compound injection port **220***a* breaks away from the mold compound **214** of the semiconductor package **200**. The breaking of the sprue from the semiconductor package **200** results in a gate cull mark **222** on the top surface **223** of the semiconductor package **100**. The gate cull mark **222** may be circular in shape. While the gate cull mark **222** of the example semiconductor package **200** is on the top surface **223**, other locations of the gate cull mark **222** are within the scope of the disclosure, the location of the gate cull mark **222** being dependent of the location of the mold compound injection port **220***a*.

[0030] FIG. **2**D is a cross section of the semiconductor package **200** of FIG. **2**C at a subsequent stage of formation during a singulation sawing process. FIG. **2**D shows an alternative singulation sawing process compared to the process referred to in FIG. **1**E. In FIG. **2**D, the semiconductor package **200** and the additional semiconductor packages **200***a* are mounted onto a saw tape ring **228**, which may also be referred to as a saw film. The semiconductor package **200** is singulated from the additional semiconductor packages **200***a* by a singulation sawing process using a singulation saw blade **230**. In FIG. **2**D, the singulation saw blade **230** is over the mold compound depression **227**. Sawing through the mold compound depression **227** results in a tapered mold compound sidewall **243** above the plurality of leads **204**, and a vertical mold compound sidewall **241** referred to in FIG. **2**E between the plurality of leads **204**. It may be advantageous for the sawing process to be centered on the mold compound depression **227** to minimize the sawing passes necessary to singulate the semiconductor package **200** from additional semiconductor packages **200***a* of the lead frame **202**. A sawing process within the mold compound depression **227** may result in a semiconductor package **200** after singulation with the tapered mold compound sidewall **243** such that the perimeter of the mold compound **214** around the top surface **223** is less than the perimeter of the mold compound **214** around the bottom surface **203**.

[0031] The singulation sawing process cuts through and separates the plurality of leads **204** of the semiconductor package **200** from the plurality of additional leads **204***a* of the additional semiconductor packages **200***a*, as well as the mold compound **214** between the semiconductor package **200** and the additional semiconductor packages **200***a*. The semiconductor package **200** and the additional semiconductor packages **200***a* are subsequently removed from the saw tape ring **228**. The singulation sawing process produces exposed side faces **232** of the plurality of leads **204** which are free of striations, striations being typical of a semiconductor package **200** formed by shearing, instead of sawing the plurality of leads **204** from plurality of additional leads **204***a* as shown in FIG. **2**D

[0032] FIG. **2**E is a perspective view of the semiconductor package **200** referred to in FIG. **2**D after singulation. The semiconductor package **200** after singulation may have a gate cull mark **222** and tapered mold compound sidewalls **243** above the plurality of leads **204**, and vertical mold compound sidewalls **241** between the plurality of leads **204**. The gate cull mark **222** may be on the exterior of the semiconductor package **200** on the top surface **223** of the mold compound **214** ash shown in FIG. **1**E, or may be on the tapered mold compound sidewall **243** (not specifically shown). Other features visible on the semiconductor package **200** after singulation include the plurality of leads **204** and exposed side faces **232** of the plurality of leads **204**, the exposed side faces being free of striations. Alternatively, the semiconductor package **200** can be singulated using the sawing process previously described above and shown in FIG. **1**E. Similarly, the semiconductor package **100** can be singulated using the sawing process described above and shown in FIG. **2**D.

[0033] While various embodiments of the present disclosure have been described above, it should be understood that they have been presented by way of example only and not limitation. Numerous changes to the disclosed embodiments can be made in accordance with the disclosure herein without departing from the spirit or scope of the disclosure. Thus, the breadth and scope of the present invention should not be limited by any of the above described embodiments. Rather, the scope of the disclosure should be defined in accordance with the following claims and their equivalents.

What is claimed is:

1. A semiconductor package, comprising:
- a plurality of leads around a perimeter of the semiconductor package;
- a microelectronic component electrically coupled to the plurality of leads;
- a mold compound contacting portions of the plurality of leads and the microelectronic component, the mold compound being electrically insulating, the mold compound having a top surface, a bottom surface, and a mold compound sidewall; and
- a gate cull mark on an exterior surface of the mold compound.

2. The semiconductor package of claim **1**, wherein the gate cull mark is on the top surface of the mold compound.

3. The semiconductor package of claim **1**, wherein the mold compound sidewall is vertical.

4. The semiconductor package of claim **1**, wherein the mold compound sidewall is tapered.

5. The semiconductor package of claim **1**, wherein the microelectronic component of the semiconductor package has wire bond electrical coupling to the plurality of leads.

6. The semiconductor package of claim **1**, wherein the microelectronic component of the semiconductor package has solder ball electrical coupling to the plurality of leads.

7. The semiconductor package of claim **1**, wherein an exposed side face of the plurality of leads, is free of striations.

8. A method of forming a semiconductor package, comprising:
- electrically coupling a microelectronic component to a plurality of leads of the semiconductor package extending to a perimeter of the semiconductor package;
- forming a mold compound on the plurality of leads and the microelectronic component, the mold compound having a top surface and bottom surface wherein:
  - a top mold chase and a middle mold chase under the top mold chase are over the microelectronic component and the plurality of leads, the top mold chase containing top mold chase runners;
  - a mold compound injection port extends through the middle mold chase and into the top mold chase and intersects the top mold chase runners, wherein the mold compound is distributed to the microelectronic component and the plurality of leads through the mold compound injection port; and
  - a bottom mold chase is under the microelectronic component and plurality of leads;
- removing the top mold chase, the middle mold chase and the bottom mold chase, the mold compound injection port forming a gate cull mark on the mold compound of the semiconductor package, and
- singulating the semiconductor package through the plurality of leads.

9. The method of claim **8**, wherein the semiconductor package is singulated between a mold compound depression and the microelectronic component of the semiconductor package.

10. The method of claim **8**, wherein the semiconductor package is singulated along a mold compound depression between the semiconductor package and an adjacent semiconductor package.

11. The method of claim **8**, wherein the middle mold chase contains middle mold chase ridges, the middle mold chase ridges forming a single mold cavity surrounding each microelectronic component and the plurality of leads of the semiconductor package.

12. The method of claim **8**, comprising forming the gate cull mark on the top surface of the mold compound.

13. The method of claim **8**, comprising forming the gate cull mark on a mold compound depression of the mold compound.

14. The method of claim **8**, comprising forming the mold compound wherein the mold compound injection port is over a center point of a mold cavity.

15. The method of claim **8**, wherein the mold compound contains filler particles, the filler particles being of silicon dioxide, aluminum oxide, or similar material.

16. The method of claim **8**, wherein the microelectronic component of the semiconductor package has wire bond electrical coupling to the plurality of leads.

17. The method of claim **8**, wherein the microelectronic component of the semiconductor package has solder ball electrical coupling to the plurality of leads.

18. The method of claim **8**, wherein the mold compound is formed on a second plurality of leads and a second microelectronic component of a second semiconductor package, wherein the top mold chase and the middle mold chase have a second mold compound injection port, wherein the mold compound is distributed to the second semiconductor package through the second mold compound injection port.

19. The method of claim **8**, comprising forming an exposed side face of the plurality of leads which is free of striations.

20. A method of forming a semiconductor packages, comprising:
- electrically coupling a plurality of microelectronic components to corresponding pluralities of leads extending to corresponding perimeters of the semiconductor packages;
- forming a mold compound on the plurality of microelectronic components and the corresponding pluralities of leads, the mold compound having a top surface and a bottom surface, wherein:
  - a top mold chase and a middle mold chase under the top mold chase are over the top surface of the mold compound, the middle mold chase containing middle mold chase ridges, the middle mold chase ridges forming a single mold cavity surrounding each of the plurality of microelectronic components and the corresponding pluralities of leads, the top mold chase containing top mold chase runners;
  - a plurality of mold compound injection ports extend through the middle mold chase and into the top mold chase and intersect with the top mold chase runners, wherein there is a mold compound injection port located over each of the semiconductor packages, wherein the mold compound is distributed to each microelectronic component and corresponding plurality of leads through a corresponding mold compound injection port; and
  - a bottom mold chase is under the plurality of microelectronic components and pluralities of leads;
- removing the top mold chase, the middle mold chase and the bottom mold chase, the mold compound injection port located over each of the semiconductor packages

forming a gate cull mark on the mold compound of each of the semiconductor packages, and

singulating the semiconductor packages by sawing through the pluralities of leads.

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